# AI-Driven Early-Warning and Decision-Support Services for Disaster-Resilient Smart Cities

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***Abstract*—Tropical cyclones pose significant risks to coastal smart cities, where timely and reliable track forecasts are critical for evacuation planning, transportation management, emergency-resource allocation, and the protection of critical infrastructure. This paper presents an AI-driven short-range tropical cyclone track forecasting framework based on a residual Graph Neural Network (GNN)–Transformer for smart city disaster resilience. The model integrates cyclone trajectories from the International Best Track Archive for Climate Stewardship (IBTrACS) with selected atmospheric variables from ERA5 reanalysis. Twelve consecutive six-hourly states are used to predict cyclone displacement at 6-, 12-, and 24-h horizons. A temporal graph convolution captures local dependencies between successive cyclone states, while Transformer blocks model longer-range temporal relationships. Analysis of an initial absolute-displacement model shows that directional error, rather than displacement magnitude alone, is the dominant source of forecasting error. To address this limitation, a persistence-guided residual learning strategy is introduced, in which recent cyclone motion provides a physically meaningful baseline and the network learns only the deviation from this trajectory. On the independent 2014 Western Pacific test set, the proposed model reduces the 24-h mean position error from approximately 225 km to 87 km and the angular error from approximately 55° to 16°. Track-level analysis further indicates that the largest remaining errors are mainly associated with pronounced recurvature events. The proposed framework demonstrates how lightweight, data-driven forecasting can support intelligent early-warning and decision-support services for disaster-resilient smart cities.**

***Index Terms*—Smart cities, tropical cyclone track forecasting, disaster resilience, artificial intelligence, graph neural networks, Transformer, residual learning, intelligent early warning**

## I. Introduction

Smart cities increasingly rely on data-driven systems for hazard monitoring, infrastructure protection, and rapid disaster response. Tropical cyclones (TCs) are a major threat to coastal cities, as track changes can disrupt evacuation, transportation, power, communication, and emergency-resource allocation [1]. Reliable short-range track forecasting is therefore important for intelligent disaster management and urban resilience. Traditional TC forecasting mainly relies on numerical weather prediction (NWP), statistical methods, and expert analysis. Although effective, NWP remains computationally intensive and sensitive to uncertainties in initial conditions and model formulation. Data-driven AI provides a complementary approach with fast inference and growing potential for large-scale weather forecasting. Pangu-Weather introduced three-dimensional neural networks for medium-range global weather prediction [2], while GraphCast employed graph neural networks to achieve skillful forecasts of global weather and severe events, including tropical cyclones [3]. FuXi further reduced long-range forecast-error accumulation through a cascaded machine-learning architecture [4]. More recently, foundation models and AI-based ensemble systems have extended data-driven weather prediction toward multiple Earth-system tasks and operational tropical cyclone forecasting [5]–[7].

In parallel, task-specific deep-learning methods have been developed for TC track forecasting. Giffard-Roisin et al. fused historical storm trajectories with aligned atmospheric reanalysis fields for 24-h track prediction [8], demonstrating the value of combining trajectory and environmental information. Transformer-based models have subsequently been introduced to capture long-range dependencies in TC time series and have shown improvements over recurrent architectures in short-term track and intensity prediction [9]. Comparative studies have also evaluated temporal convolutional networks, LSTM, Transformer, and related architectures, emphasizing that prediction strategy and input representation can be as important as the choice of network itself [10]. More recently, TropiCycloneNet combined a large multimodal global TC dataset with a meteorology-informed deep-learning framework, highlighting the importance of jointly representing storm history and environmental context [11]. These developments show a transition from purely trajectory-based learning toward models integrating atmospheric information and stronger spatiotemporal representations.

Despite these advances, two limitations remain important for AI-based short-range cyclone forecasting intended for intelligent early-warning applications. First, geographical position error alone provides limited insight into the source of a failed forecast. In particular, it does not distinguish whether the error arises from an incorrect displacement magnitude or an incorrect direction of motion. Such distinction is important for disaster decision support because directional errors during turning or recurvature events can lead to substantially different affected areas. Second, direct prediction of the complete future displacement requires the learning model to reproduce a large component of cyclone motion that is already encoded in the most recent trajectory. This motivates a hybrid formulation in which recent motion is used as a physically meaningful prior,

while the neural network focuses on learning deviations from that prior.

To address this, we propose a residual GNN–Transformer framework for short-range tropical cyclone forecasting. Cyclone history is modeled as a temporal graph, where graph convolution captures local dependencies and Transformer blocks learn longer-range temporal relationships. The model combines IBTrACS cyclone tracks with selected ERA5 atmospheric variables and adopts a persistence-guided residual formulation, in which the network learns only the correction to the recent-motion baseline. The resulting framework is compact and suitable for intelligent early-warning and smart-city disaster decision-support systems. The model combines cyclone-track information from the International Best Track Archive for Climate Stewardship (IBTrACS) [12] with atmospheric variables derived from ERA5 reanalysis [13].

The main contributions of this work are threefold:

1) A temporal GNN-Transformer architecture is developed to jointly model recent cyclone dynamics and atmospheric information for short-range 6-, 12-, and 24-h track forecasting.
2) A direction-magnitude error decomposition is introduced to diagnose the dominant failure mode of the initial forecasting model, showing that directional error is the primary performance bottleneck.
3) A persistence-guided residual learning strategy is proposed to incorporate recent cyclone motion as a physically meaningful prior, reducing the 24-h mean position error from approximately 225 km to 87 km and the angular error from 55° to 16° on the independent 2014 test set.

These contributions provide an AI-based forecasting component that can support intelligent early-warning and disaster decision-support services for coastal smart cities.

## II. Forcasting Data and Problem Formulation

### A. Cyclone Track and Atmospheric Data

Two complementary data sources are used in this study: cyclone-track observations from IBTrACS [12](`IBTrACS.WP.v04r01.nc`) and atmospheric variables from ERA5 reanalysis [13]. The Western Pacific subset of IBTrACS provides latitude–longitude positions at 6-h intervals, while ERA5 describes the large-scale atmospheric environment affecting cyclone motion. Data from 2004–2012 are used for training, 2013 for validation, and 2014 as an independent test set. This chronological split provides a realistic evaluation of generalization to future storms.

The atmospheric environment is described using ERA5 reanalysis [13]. Annual ERA5 fields from 2004 to 2014 are considered over the domain $0°$–$40°$N and $100°$–$180°$E, covering the main region of interest in the Western Pacific. ERA5 is used here because cyclone motion is strongly influenced by the surrounding large-scale atmospheric circulation, which cannot be described from the track coordinates alone.

The data are divided chronologically rather than randomly. Storms from 2004–2012 are used for training, 2013 is reserved for validation, and 2014 is kept as an independent test set. This separation is important for a forecasting problem because a random split could place observations from the same period, or closely related atmospheric conditions, in both the training and test sets. The chronological split therefore provides a more realistic assessment of generalization to future storms.

TABLE I
Input Features Used at Each Time Step

| Group | Variables |
|---|---|
| Position | $\phi, \lambda$ |
| Motion | $\Delta\phi, \Delta\lambda$ |
| Dynamics | speed, $\sin(\alpha)$, $\cos(\alpha)$ |
| ERA5 | $u_{200}, v_{200}, u_{500}, u_{850}, vo_{850}, v_{500}, z_{500}$ |
| Steering flow | $steer_u, steer_v$ |

### B. Feature Selection and Representation

The initial ERA5 feature pool contains 17 atmospheric variables sampled over 236,352 cyclone records. It includes zonal and meridional winds at 200, 500, and 850 hPa, relative vorticity at 850 hPa, geopotential fields, temperature, humidity, mean sea-level pressure, sea-surface temperature, and 10-m winds. Rather than feeding all available variables to the network, we first examined their redundancy and predictive relevance.

The selection was carried out in several steps. Pearson correlation was used to identify strongly correlated variable pairs, while the variance inflation factor (VIF) was used to detect multivariate collinearity. Principal component analysis provided an additional view of the overall redundancy of the candidate set. Finally, Random Forest importance and SHAP values were used to examine the contribution of individual variables to displacement prediction. This procedure retained six dynamic ERA5 variables:

$$u_{200},\ v_{200},\ u_{500},\ u_{850},\ vo_{850},\ v_{500}.$$

Although $z_{500}$ was initially removed by the statistical selection, it was subsequently reintroduced because the 500-hPa geopotential field contains physically relevant information on the large-scale steering environment. This resulted in seven ERA5 variables used by the final model.

The atmospheric variables are combined with information derived directly from the cyclone trajectory. Table I summarizes the final 16-dimensional feature vector at each time step. Local ERA5 variables are extracted within a $1°$ radius around the cyclone center, whereas the steering-flow components are averaged over a larger $4°$ radius to represent the surrounding flow rather than the circulation immediately around the vortex.

The heading angle $\alpha$ is represented by its sine and cosine rather than by the angle itself, avoiding the discontinuity at $\pm\pi$. Before training, each feature $x$ is scaled using the same affine normalization adopted throughout the experiments,

$$\tilde{x} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}. \tag{1}$$

### C. Multi-Horizon Forecasting

At a given forecast time $t$, the model receives the most recent 12 six-hourly steps. Since consecutive observations are separated by 6 h, this corresponds to a 72-h history. Each time step contains the 16 features listed in Table I; hence, one input sample can be written as

$$\mathbf{X}_t = [\mathbf{x}_{t-11}, \ldots, \mathbf{x}_t] \in \mathbb{R}^{12\times 16}, \qquad \mathbf{x}_i \in \mathbb{R}^{16}. \tag{2}$$

The prediction target is expressed as cyclone displacement rather than absolute geographical position. For a forecast horizon $h$, the true displacement from the current position is defined as

$$\mathbf{d}_h = \begin{bmatrix} \phi_{t+h} - \phi_t \\ \lambda_{t+h} - \lambda_t \end{bmatrix}, \qquad h \in \{1, 2, 4\}, \tag{3}$$

where one step corresponds to 6 h. The three output horizons are therefore +6, +12, and +24 h:

$$\hat{\mathbf{d}}_{6h}, \qquad \hat{\mathbf{d}}_{12h}, \qquad \hat{\mathbf{d}}_{24h}. \tag{4}$$

Predicting displacement instead of absolute position reduces the dependence of the learning problem on the geographical coordinates of the basin. For the error diagnosis, we also consider the displacement magnitude and direction. Let $\mathbf{d}$ and $\hat{\mathbf{d}}$ denote the observed and predicted displacement vectors, respectively. The speed ratio and directional error are defined as:

$$s = \frac{\|\hat{\mathbf{d}}\|}{\|\mathbf{d}\|}, \qquad \theta = \left|\operatorname{atan2}\left(\sin(\hat{\alpha} - \alpha), \cos(\hat{\alpha} - \alpha)\right)\right|. \tag{5}$$

where $s < 1$ indicates that the predicted displacement is too short, $\alpha$ and $\hat{\alpha}$ are the observed and predicted motion directions. The value of $\theta$ is reported in degrees. These two measures are used together with the position error to identify whether a forecast error mainly comes from displacement magnitude or direction. This diagnostic provides a natural starting point for the persistence-guided residual formulation introduced in Section III.

## III. RESIDUAL GNN–TRANSFORMER FORECASTING FRAMEWORK

The proposed framework combines temporal graph learning, Transformer-based sequence modeling, and persistence-guided residual prediction. The temporal GNN captures local relationships between successive cyclone states, while the Transformer models longer-range dependencies over the observation window. Instead of predicting the complete future displacement directly, the network learns the residual correction to a persistence-based motion estimate. Fig. 1 illustrates the overall architecture.

### A. Temporal Graph Modeling

The input described in Section II contains 12 time steps, with 16 features at each step. In this work, the graph is a temporal graph, not a spatial graph. Each node represents one time step of the cyclone history. A node is connected to its three previous and three following neighbors when they are

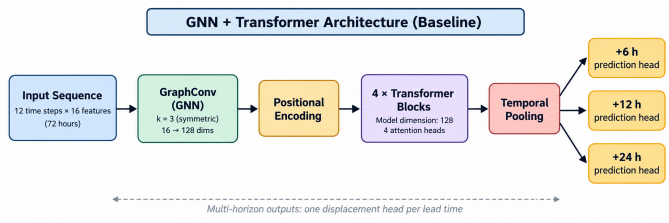



Fig. 1. Architecture of the proposed temporal GNN–Transformer for multi-horizon tropical cyclone track forecasting.

available. Thus, the graph contains information from the same 72-h history as the input sequence.

Let $\mathbf{A}$ denote the adjacency matrix after adding self connections, and let $\mathbf{D}$ be its degree matrix. We use the normalized adjacency matrix as in the GCN formulation [14]:

$$\hat{\mathbf{A}} = \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}. \tag{6}$$

The input tensor has size $B \times 12 \times 16$, where $B$ is the batch size. A dense projection first changes the feature dimension from 16 to 128. The graph operation then mixes the information of neighboring time steps. In a simple form, this operation can be written as

$$\mathbf{H}^{(1)} = \sigma\left(\hat{\mathbf{A}}\mathbf{X}\mathbf{W}_g + \mathbf{b}_g\right), \tag{7}$$

where $\mathbf{W}_g$ and $\mathbf{b}_g$ are trainable parameters and $\sigma(\cdot)$ denotes the activation function. The output of this block has size $B \times 12 \times 128$.

### B. Transformer-Based Temporal Learning

While graph convolution captures local temporal dependencies, Transformer blocks model relationships across the complete 72-h sequence. Sinusoidal positional encoding preserves temporal order, followed by four multi-head self-attention blocks. The representation of the most recent cyclone state is then used by separate prediction heads for the +6, +12, and +24-h horizons. Each block has four attention heads, with a dimension of 32 for each head. The usual scaled dot product attention [15] is used:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V}, \tag{8}$$

where $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are the query, key, and value matrices, respectively, and $d_k = 32$ in our implementation.

After the attention operation, each Transformer block contains a residual connection, layer normalization, and a feed forward network with dimensions $128 \to 512 \to 128$. The output dimension therefore remains $B \times 12 \times 128$ after each block.

After the fourth Transformer block, only the representation of the last time step is retained. We found this choice more suitable than averaging all time steps because the last state contains the most recent motion of the cyclone. The resulting 128 dimensional vector passes through two dense layers and

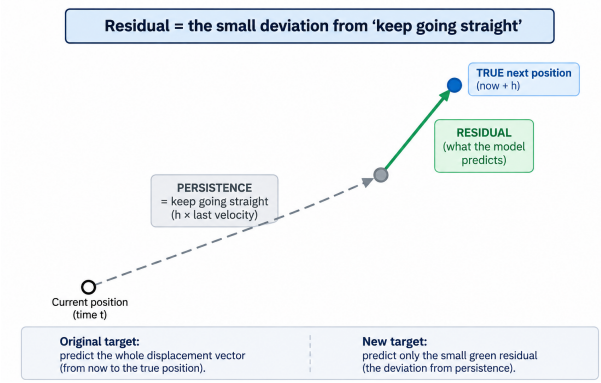



Fig. 2. Principle of the residual formulation. The model predicts the difference between the actual displacement and the persistence displacement.

is then sent to three separate prediction heads for 6, 12, and 24 h.

### C. Persistence-Guided Residual Learning

Our first version of the model directly predicted the future displacement. The experiments showed that this formulation had a large direction error. In particular, changing only the predicted displacement magnitude did not solve the problem. We therefore changed the target used for training.

A simple persistence prediction assumes that the cyclone continues with its latest observed displacement. If $\mathbf{v}_{\text{last}}$ is the displacement during the last 6-h interval, the persistence estimate for horizon $h$ is

$$\hat{\mathbf{d}}_h^{\text{pers}} = h\,\mathbf{v}_{\text{last}}, \qquad h \in \{1, 2, 4\}. \tag{9}$$

Here, $h = 1$, 2, and 4 correspond to 6, 12, and 24 h, respectively. Instead of predicting $\mathbf{d}_h$ directly, the network is trained to predict the difference between the true displacement and this persistence estimate:

$$\mathbf{r}_h = \mathbf{d}_h - h\,\mathbf{v}_{\text{last}}. \tag{10}$$

The idea is illustrated in Fig. 2. The persistence part already gives a simple estimate of where the cyclone will move. The neural network therefore has a smaller task: it learns how the real trajectory deviates from this estimate.

During prediction, the network gives the estimated residual $\hat{\mathbf{r}}_h$. The final displacement is then obtained by adding the persistence component back:

$$\hat{\mathbf{d}}_h = h\,\mathbf{v}_{\text{last}} + \hat{\mathbf{r}}_h. \tag{11}$$

An important point is that a zero residual simply gives the persistence prediction. The network only needs to learn the correction when the cyclone starts to depart from its recent motion. This formulation is especially useful when the direction begins to change.

TABLE II
COMPARISON OF PERSISTENCE AND THE INITIAL GNN–TRANSFORMER ON THE 2014 TEST SET

| Method / Horizon | Error (km) | Speed Ratio | Angle Error |
|---|---|---|---|
| Persistence, +6 h | 39.6 | 1.00 | 12.6° |
| Persistence, +12 h | 87.9 | 1.00 | 14.6° |
| Persistence, +24 h | 208.1 | 1.00 | 18.5° |
| Initial model, +6 h | 58.2 | 0.67 | 57° |
| Initial model, +12 h | 114.9 | 0.80 | 56° |
| Initial model, +24 h | 225.2 | 0.77 | 55° |

### D. Training and Prediction Procedure

The model is trained with a weighted mean absolute error. For one prediction head, the loss is

$$\mathcal{L}_h = \frac{1}{N}\sum_{i=1}^{N}\left[w_\phi\,|\hat{r}_{\phi,i} - r_{\phi,i}| + w_\lambda\,|\hat{r}_{\lambda,i} - r_{\lambda,i}|\right], \tag{12}$$

where $w_\phi = 1.0$ and $w_\lambda = 0.5$. The three forecast horizons are combined using weights 0.8, 1.0, and 1.3 for 6, 12, and 24 h, respectively. The larger weight for the 24-h head gives more importance to the more difficult forecast horizon.

The model was trained using Adam with a learning rate of $5 \times 10^{-3}$ and a dropout rate of 0.2. The complete model contains approximately $8.2 \times 10^5$ trainable parameters. The total loss is

$$\mathcal{L} = 0.8\mathcal{L}_{6h} + 1.0\mathcal{L}_{12h} + 1.3\mathcal{L}_{24h}. \tag{13}$$

## IV. Experimental Results and Discussion

All results in this section are obtained on the independent 2014 test set. Unless otherwise stated, the shadow evaluation protocol is used. In this protocol, each forecast starts from an observed cyclone position and the predictions are not chained. This setting is close to the practical case where a new cyclone position becomes available every six hours.

### A. Baseline Comparison and Directional Diagnosis

We first compare the initial GNN–Transformer with a simple persistence model. This comparison was useful because the first model gave reasonable position errors, but it was not clear where the remaining error came from. The results are given in Table II.

The initial model does not outperform persistence. More importantly, Table II shows that the main difference comes from the predicted direction. The angular error of the initial model remains between 55° and 57°, compared with only 12.6°–18.5° for persistence. The speed ratio is also below one, indicating some underestimation of the displacement magnitude, but this alone cannot explain the large position error.

Figure 3 illustrates this problem geometrically. When the predicted displacement has a large angular error, changing its magnitude does not correct its orientation. The end point therefore remains far from the observed position even if the vector length is adjusted. This observation suggests that the main problem is not simply how far the cyclone moves, but in

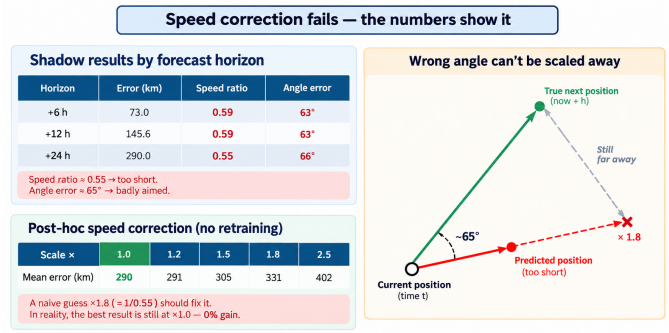



Fig. 3. Directional error diagnosis. Changing the magnitude of a predicted displacement cannot correct a large error in its direction.

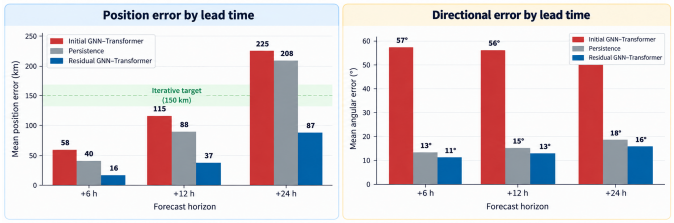



Fig. 4. Position error (left) and angular error (right) at different forecast horizons for the initial GNN–Transformer, persistence, and the residual GNN–Transformer.

which direction it moves. This diagnosis motivated the change of prediction target introduced in Section III-C.

## B. Effect of Residual Learning

After the above diagnosis, the model was trained again using the residual target defined in Section III. Figure 4 compares the initial model, persistence, and the residual model at the three forecast horizons.

The improvement is clear at all three horizons. At +24 h, the mean position error decreases from 225 km to 87 km, corresponding to a reduction of about 61%. The angular error also decreases from $55^\circ$ to $16^\circ$. The residual model therefore outperforms both the initial model and persistence in position and direction.

The simultaneous reduction of the two errors supports the diagnosis in the previous subsection. The improvement does not come only from changing the displacement magnitude. By using recent cyclone motion as the persistence reference, the network learns the correction needed to depart from this motion.

## C. Track Level Analysis

The average errors do not tell the whole story. We also inspected individual cyclone tracks to understand when the model works and when it fails. Figure 5 shows examples from the earlier model after reintroducing the 500-hPa geopotential, $z_{500}$. This figure is included here as a diagnostic result rather than as a result of the final residual model.

For relatively regular tracks, the predicted trajectory follows the general motion of the cyclone. A different behavior appears

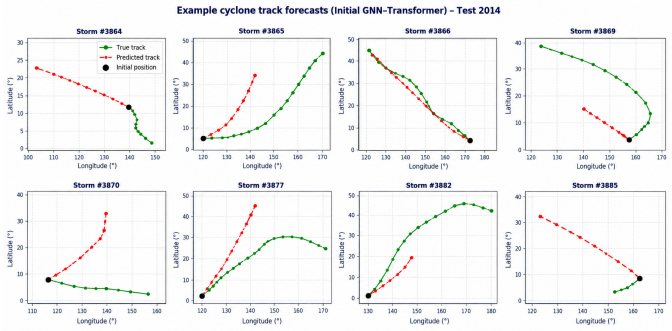



Fig. 5. Observed (green) and predicted (red) cyclone tracks after reintroducing the 500-hPa geopotential variable. The model captures the general motion but remains too straight in several recurvature cases.

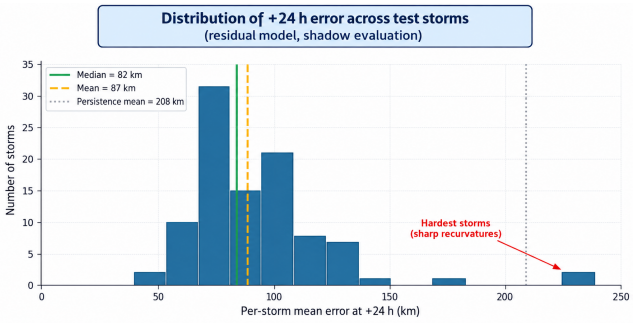



Fig. 6. Distribution of the mean +24-h position error across storms for the residual model.

when the cyclone starts to turn. In the recurvature cases, particularly storms 3865, 3869, and 3877 in the test analysis, the predicted track tends to continue almost straight while the observed cyclone changes direction. Thus, adding $z_{500}$ helped the model to capture the general direction, but it was not enough to describe some strong turns.

The same behavior can still be seen, in a different way, after residual training. Figure 6 gives the distribution of the mean +24-h error for individual storms. The median is clearly below the mean. In other words, most storms have moderate errors, while a small number of difficult storms increase the average error considerably. Inspection of these cases in the report showed that many of the large errors correspond to pronounced recurvature.

## D. Discussion

The results support the two main methodological contributions of this work. First, separating the forecast error into direction and magnitude gives a clearer diagnosis than position error alone. The initial model has an angular error of about $55^\circ$–$57^\circ$, while changing the displacement magnitude does not improve the forecast. This confirms that direction is the main limitation of the initial formulation.

Second, persistence is used not only as a baseline, but also as a motion reference for learning. After replacing the absolute displacement target by the residual from persistence, the +24-h position error decreases from 225 km to 87 km, while the

TABLE III
COMPARISON WITH RELATED DEEP LEARNING APPROACHES FOR TROPICAL CYCLONE TRACK FORECASTING

| Study | Main model | Input information | Region | 24-h result | Main focus |
|---|---|---|---|---|---|
| Giffard-Roisin *et al.* [8] | Fusion CNN | Track + reanalysis fields | WP / global | 136.1 km (WP) | Fusion of trajectory and atmospheric fields |
| Jiang *et al.* [9] | Transformer | TC track and intensity attributes | NWP | – | Temporal attention and feature augmentation |
| Gan *et al.* [10] | LSTM, ConvLSTM, TCN, Transformer | Historical TC attributes | NWP | – | Comparison of models and prediction settings |
| Huang *et al.* [11] | TropiCycloneNet | Track + meteorological fields + environmental data | Global | – | Multimodal and meteorological information |
| **This work** | **Residual GNN–Transformer** | **Track + selected ERA5 variables** | **WP** | **87 km** | **Direction diagnosis + persistence residual** |

*Note:* The reported errors are not directly comparable because the studies use different datasets, test periods, input variables, and evaluation protocols. The table is intended to show the methodological context rather than provide a direct ranking of model performance.

angular error decreases from $55^\circ$ to $16^\circ$. The simultaneous improvement of both metrics supports the proposed residual formulation.

Table III places these results in the context of recent deep learning approaches. Previous studies have explored atmospheric data fusion [8], Transformer based prediction [9], comparisons of temporal models and prediction strategies [10], and multimodal meteorological information [11]. In contrast, the focus of this work is on diagnosing the directional error and then reformulating the learning target using persistence. Therefore, the main contribution is not simply the GNN–Transformer architecture, but the sequence from error diagnosis to a physically meaningful residual prediction. The comparison should not be interpreted as a direct performance ranking because the studies use different datasets and evaluation protocols.

## V. CONCLUSION

This paper presented a residual GNN–Transformer for short-range tropical cyclone track forecasting in the Western Pacific. Error analysis showed that directional error, rather than displacement magnitude, was the main limitation of the initial model. By using recent cyclone motion as a persistence reference and learning only the residual correction, the proposed model reduced the +24-h mean position error from about 225 km to 87 km and the angular error from $55^\circ$ to $16^\circ$ on the independent 2014 test set. The remaining large errors were mainly associated with strong recurvature events. These results demonstrate the potential of the proposed framework for intelligent early-warning and disaster decision-support systems in coastal smart cities. Future work will incorporate spatial ERA5 fields to better capture atmospheric structures associated with changes in cyclone direction.

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## REFERENCES

[1] A. J. Garner, D. Samanta, M. M. Weaver, *et al.*, "Changes to tropical cyclone trajectories in Southeast Asia under a warming climate," *npj Climate and Atmospheric Science*, vol. 7, Art. no. 156, Jul. 2024.

[2] K. Bi, L. Xie, H. Zhang, X. Chen, X. Gu, and Q. Tian, "Accurate medium-range global weather forecasting with 3D neural networks," *Nature*, vol. 619, pp. 533–538, Jul. 2023.

[3] R. Lam, A. Sanchez-Gonzalez, M. Willson, *et al.*, "Learning skillful medium-range global weather forecasting," *Science*, vol. 382, no. 6677, pp. 1416–1421, Dec. 2023.

[4] L. Chen, X. Zhong, F. Zhang, Y. Cheng, Y. Xu, Y. Qi, and H. Li, "FuXi: A cascade machine learning forecasting system for 15-day global weather forecast," *npj Climate and Atmospheric Science*, vol. 6, Art. no. 190, Nov. 2023.

[5] C.-C. Liu, K. Hsu, M. S. Peng, *et al.*, "Evaluation of five global AI models for predicting weather in Eastern Asia and Western Pacific," *npj Climate and Atmospheric Science*, vol. 7, Art. no. 221, Sep. 2024.

[6] C. Bodnar, W. P. Bruinsma, A. Lucic, *et al.*, "A foundation model for the Earth system," *Nature*, vol. 641, pp. 1180–1187, May 2025.

[7] F. Alet, T. R. Andersson, I. Price, *et al.*, "Operational tropical cyclone forecasting with AI," *Nature*, 2026, doi: 10.1038/s41586-026-10953-2.

[8] S. Giffard-Roisin, M. Yang, G. Charpiat, C. Kumler Bonfanti, B. Kégl, and C. Monteleoni, "Tropical cyclone track forecasting using fused deep learning from aligned reanalysis data," *Frontiers in Big Data*, vol. 3, Art. no. 1, Jan. 2020.

[9] W. Jiang, D. Zhang, G. Hu, T. Wu, L. Liu, Y. Xiao, and Z. Duan, "Transformer-based tropical cyclone track and intensity forecasting," *Journal of Wind Engineering and Industrial Aerodynamics*, vol. 238, Art. no. 105440, Jul. 2023.

[10] S. L. Gan, J. Y. Fu, G. F. Zhao, P. W. Chan, and Y. C. He, "Short-term prediction of tropical cyclone track and intensity via four mainstream deep learning techniques," *Journal of Wind Engineering and Industrial Aerodynamics*, vol. 244, Art. no. 105633, Jan. 2024.

[11] C. Huang, P. Mu, J. Zhang, S. Chan, S. Zhang, H. Yan, S. Chen, and C. Bai, "Benchmark dataset and deep learning method for global tropical cyclone forecasting," *Nature Communications*, vol. 16, Art. no. 5923, Jul. 2025.

[12] K. R. Knapp, M. C. Kruk, D. H. Levinson, H. J. Diamond, and C. J. Neumann, "The International Best Track Archive for Climate Stewardship (IBTrACS): Unifying tropical cyclone data," *Bulletin of the American Meteorological Society*, vol. 91, no. 3, pp. 363–376, Mar. 2010.

[13] H. Hersbach, B. Bell, P. Berrisford, *et al.*, "The ERA5 global reanalysis," *Quarterly Journal of the Royal Meteorological Society*, vol. 146, no. 730, pp. 1999–2049, Jul. 2020.

[14] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2017.

[15] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, pp. 5998–6008, 2017.